MINING DEALER BEST PRACTICE

3500 Engine Cylinder Liner Clamp Tool

Maintenance and Repair	Application	Component Life Management	**Component Renewal (CRC)**	MARC Management

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1.0 Introduction

Installing cylinder liner and seals into the engine block is an important operation. Any issues with the proper positioning of the seals can lead to future head joint coolant leakage. Some dealers have a concern that once liners and seals are installed in the block, the liner may move during block rotation as pistons, rod, crankshaft are installed.
Some dealers have fabricated a device to hold the cylinder liner in place during short block assembly.

2.0 Best Practice Description

While most dealers are using bolts and washers to lock the liner down, Finning Chile is using a more efficient and simple device consisting of a fabricated polyurethane block and two bolts and washers. Each block clamps two liners.

Photos below show design and use of the tool.

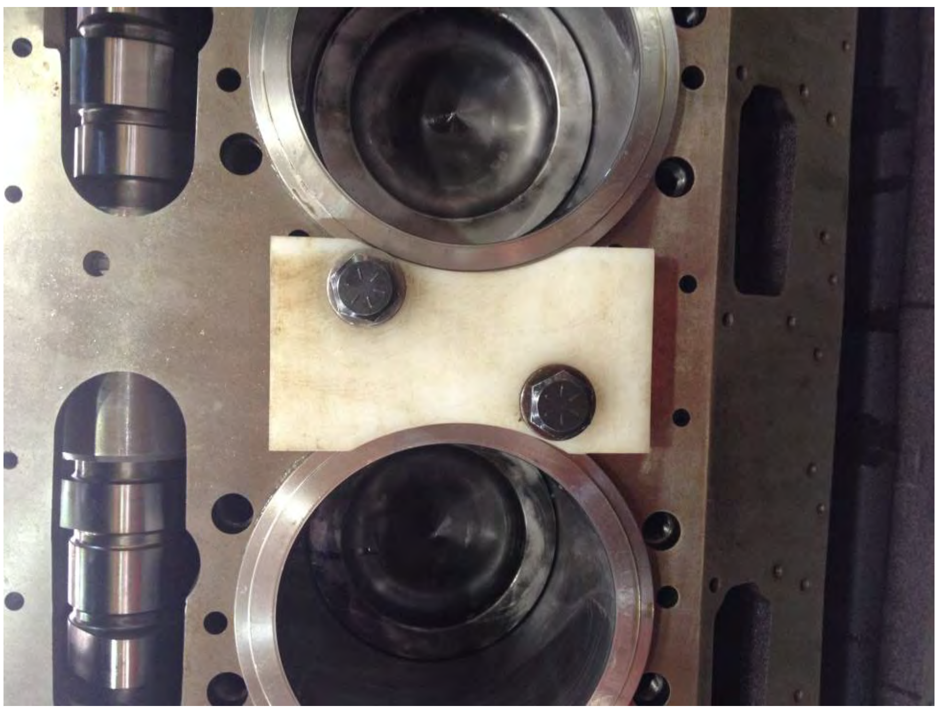

Figure 1 Clamping device and location on block. 1 clamp per 2 cylinders

Figure 2 One clamp for every 2 cylinders

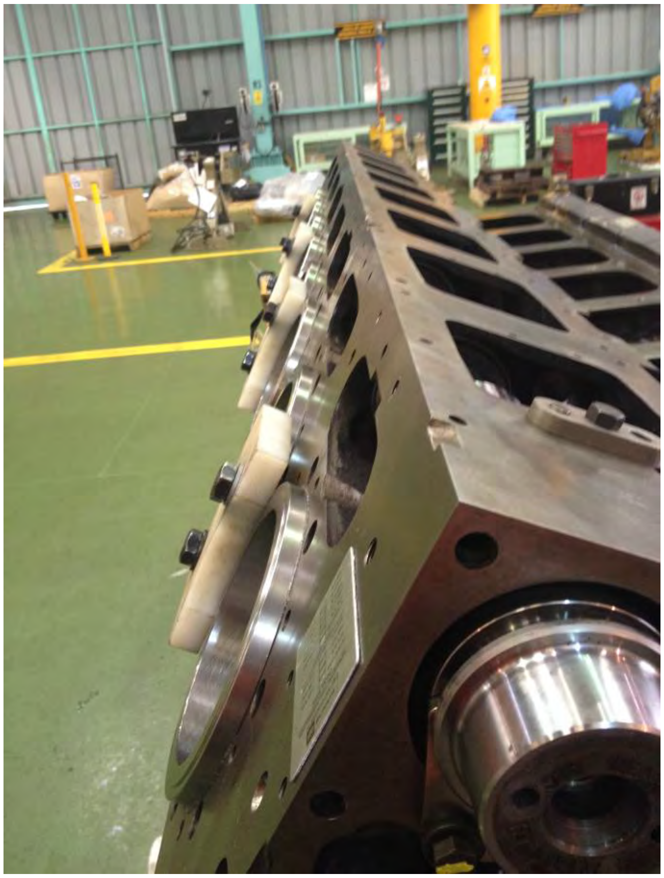

Figure 3 Side view for thickness

Figure 4 Close up view

Figure 5 One clamp for every other cylinder

3.0 Implementation Steps

Fabricate a sufficient number of polyurethane blocks cut and sized to match 3500 liner flange. 35-40 ft-lb is the bolt torque.

4.0 Benefits

- Ensures positioning of liner will not move during engine rotation.
- Protects integrity of the liner band seals.
- Reduces risk of future seal leakage.

5.0 Resources Required

- Low cost – easy to make device.

6.0 Supporting Attachments / References

- Drawings.pdf

7.0 Related Best Practices

N/A

8.0 Acknowledgements

Photos are from Finning Chile.

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